

Decomposable Performance Accounting for Agent Design Decisions

Updated Empirical Revision

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April 24, 2026

Abstract

Benchmark racing conflates per-step cost, within-mode competence, information structure, and routing quality into a single opaque comparison. The original paper showed that, under a packed latent-mode family assumption, the performance gap between two agent designs decomposes exactly into four terms: cost, competence, information gain, and routing mismatch. This revision keeps the same formalism and updates the empirical section with a real verifiable benchmark based on oracle-family workload modes.

The updated empirical picture is sharper than the original synthetic-only story. On a real query-engine optimization benchmark, the decomposition diagnoses a threshold-sensitive transition: under permissive success definitions, routing is trivial; under stricter verified-improvement thresholds, routing is non-trivial on both pilot and holdout. Under uniform task-mode priors and a 5% relative-improvement success event, the cost–competence tradeoff is strongly separated ($\log \kappa_0/\kappa = -1.424$, $\phi = 6.817$), the holdout information term is stable ($G = 0.660$, 95% bootstrap CI $[0.635, 0.686]$), and the pilot router is already much closer to the oracle than any single global model ($\varepsilon_{\text{pilot}} = 0.033$ versus $\varepsilon_{\text{single}} = 6.959$). The real-benchmark evidence therefore supports a refined claim: the decomposition is a practical diagnostic for distinguishing routing-trivial and routing-non-trivial regimes, and in the non-trivial regime it provides actionable model-selection guidance that plain benchmark racing obscures.

1 Introduction

A builder of agent systems faces a deployment question every cycle: given a task family, a compute budget, and a menu of models ranging from cheap fast systems to expensive frontier systems, which design should be used?

The dominant answer is benchmark racing: run multiple systems on a held-out benchmark, compare aggregate scores, and pick the winner. Benchmark racing is fast and often useful, but it is structurally blind. It mixes four distinct effects into one score:

1. per-step cost,
2. within-mode competence,
3. task-relevant information structure, and
4. routing mismatch.

The original paper introduced a decomposition that separates these effects exactly under a packed latent-mode family assumption. This revision updates the paper where the old manuscript

was weakest: empirical validation on a real benchmark. The key outcome is not that every realistic benchmark is automatically routing-heterogeneous. Rather, the decomposition reveals when a benchmark is routing-trivial and when it becomes genuinely mode-conditional once success is defined in a task-appropriate way.

Updated empirical claim. On the oracle-family query-engine benchmark, the decomposition reveals a threshold-sensitive transition from cost-dominated routing triviality to genuine mode-conditional model preference. At threshold $\tau = 0.00$ (any positive improvement), routing is trivial on pilot and holdout. At $\tau \in \{0.05, 0.10\}$, routing is non-trivial on pilot and holdout: the small model remains preferable on `range_local_scans`, while the larger model is necessary on `topk_stress`. In this non-trivial regime, the pilot router is already much closer to the oracle than any single global model.

2 Preliminaries

We work with externally verifiable budgeted transductive tasks. Each instance x admits a verifier $v(x, y) \in \{0, 1\}$ and the agent must produce a verified solution within a resource budget. Under the packed latent-mode family assumption, each instance has a dominant latent mode $S(x) \in \{1, \dots, M_n\}$ with prior $\pi_s = P(S = s)$, and under model M the time to verified solution is geometric with success probability $p(s, M)$ per attempt.

For the real benchmark used in this revision, the task modes are oracle workload families:

- `range_local_scans`
- `topk_stress`

An *attempt* is a one-step batched optimization run of the query-engine solver. The solver itself still uses the six action modes from the repository (`layout`, `indexing`, `topk`, `caching`, `summaries`, `micro`), but the paper-level comparison is over task modes and model choices.

3 The Decomposition Engine

3.1 Routing-Information Identity

Let D denote the routing-side-information variable available to the design, let $\pi_D^M(s) = P(S = s \mid D, M)$ be the posterior over task modes under model M , and let q_D be the routing allocation actually used after observing D . Define the log-performance functional

$$\mathcal{J}(M, q) = -\log \kappa(M) + \mathbb{E}[-\log T(M, q)].$$

Under the packed-family assumption, the general routing-information identity is:

Theorem 1 (Routing-Information Identity). *Assume the packed latent-mode family, positivity $p(s, M) \in (0, 1]$, and measurability of $D \mapsto q_D$. For two feasible designs (M_0, q_0) and (M, q) on the same task distribution, the log-performance gap*

$$\Delta(M_0, q_0 \rightarrow M, q) = \mathcal{J}(M, q) - \mathcal{J}(M_0, q_0)$$

admits the exact decomposition

$$\begin{aligned} \Delta = & \log \frac{\kappa_0}{\kappa(M)} + \sum_s \pi_s \log \frac{p(s, M)}{p(s, M_0)} \\ & + (I_\pi(S; D_M) - I_\pi(S; D_{M_0})) - \left(\mathbb{E}_{D_M} [\text{KL}(\pi_D^M \parallel q_D)] - \mathbb{E}_{D_{M_0}} [\text{KL}(\pi_D^{M_0} \parallel q_{0,D})] \right). \end{aligned} \quad (1)$$

The model-selection theorem used in the experiments is the baseline-ignorant specialization of Theorem 1. If the reference design is mode-agnostic, i.e. it uses no side information and routes with the prior $q_0(s) = \pi_s$, then $I_\pi(S; D_{M_0}) = 0$ and $\mathbb{E}_{D_{M_0}} [\text{KL}(\pi_D^{M_0} \parallel q_{0,D})] = 0$. In that case (1) reduces to the four-term formula:

Theorem 2 (Four-Term Model-Aware Decomposition). *Under the packed latent-mode family, geometric trials, and a mode-agnostic reference design ($M_0, q_0 = \pi$), the log-performance gap between the reference and a design (M, q) is*

$$\Delta = \log \frac{\kappa_0}{\kappa(M)} + \sum_s \pi_s \log \frac{p(s, M)}{p(s, M_0)} + I_\pi(S; D_M) - \mathbb{E}_D [\text{KL}(\pi_D^M \parallel q_D)]. \quad (2)$$

This is the same four-term decomposition as in the original manuscript:

- **cost:** $\log \kappa_0 / \kappa(M)$
- **competence:** $\phi = \sum_s \pi_s \log \frac{p(s, M)}{p(s, M_0)}$
- **information gain:** $G = I_\pi(S; D_M)$
- **routing mismatch:** $\varepsilon = \mathbb{E}_D [\text{KL}(\pi_D^M \parallel q_D)]$

The decomposition is diagnostic rather than dual: it measures which lever is responsible for performance differences, but it is not a shadow-price decomposition of a Lagrangian relaxation. Appendix A contains the full proof of Theorem 1 and the specialization to Theorem 2.

3.2 Crossover

For a smaller cheaper model with success probability p_s and a larger more expensive model with success probability p_ℓ , the cheap model matches one large-model attempt at retry depth

$$D_{\text{cross}} = \frac{\log(1 - p_\ell)}{\log(1 - p_s)}. \quad (3)$$

When D_{cross} lies below the cost ratio $R = \kappa_\ell / \kappa_s$, the smaller model with retries is preferable.

3.3 Modewise Selection Rule

For a fixed task mode s , the decomposition implies that the relevant per-mode score is

$$\rho(s, M) = -\log p(s, M) + \log \kappa(M).$$

The preferred model on mode s is therefore

$$M^\star(s) = \arg \min_M \rho(s, M).$$

This is exactly the quantity estimated in the oracle-family experiments: the paper does not select a model by global average score alone, but by checking which model minimizes the cost-adjusted negative log-success on each task mode.

4 Oracle-Family Benchmark Validation

4.1 Protocol

The updated validation uses the repository’s verifiable stateful query-engine benchmark. Each run is restricted to a single oracle workload family, so task mode identity is controlled by construction. We evaluate two real models: `gpt-5.3-codex-spark` and `gpt-5.3-codex`. We report the one-step setting because it is the cleanest approximation to the geometric attempt model assumed by the theory.

The main success event is:

$$\text{success} \iff \text{relative_improvement} \geq \tau,$$

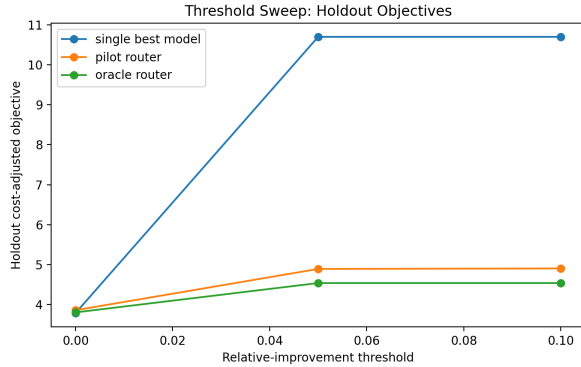
where relative improvement is measured against the verified baseline loss. For the main paper figures we use uniform task-mode priors over the two oracle families, so the decomposition is not biased by temporary sampling imbalance in the collected run matrix.

4.2 Threshold Transition

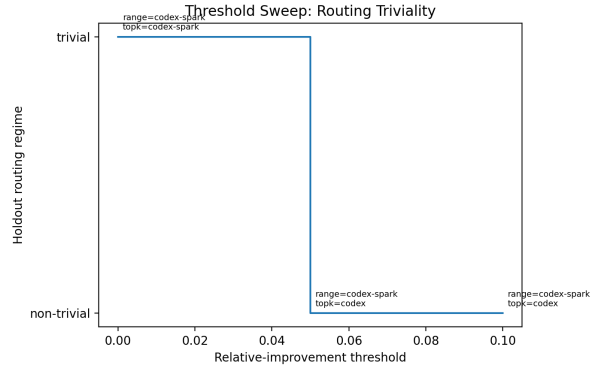
The first empirical question is whether the benchmark is routing-trivial or routing-non-trivial as the success definition tightens. Figure 1 shows a clear transition:

- at $\tau = 0.00$, routing is trivial on pilot and holdout;
- at $\tau = 0.05$ and $\tau = 0.10$, routing is non-trivial on pilot and holdout.

This validates the central diagnostic role of the decomposition: benchmark racing is sufficient in the permissive regime, but not once the success event requires a non-trivial verified gain.



(a) Holdout objectives across thresholds.



(b) Routing triviality transition.

Figure 1: Threshold sweep on the oracle-family benchmark using uniform task-mode priors. The holdout regime is trivial at $\tau = 0$ and non-trivial for $\tau \geq 0.05$.

4.3 Four-Term Diagnosis at $\tau = 0.05$

Table 1 gives the main decomposition quantities at the 5% threshold, with 95% bootstrap intervals. The pairwise cost term is sharply negative, reflecting the $\sim 4.15\times$ cost ratio in favor of the smaller model. The competence term is sharply positive, reflecting the fact that the larger model is necessary on `topk_stress`. The holdout information gain is stable and strictly positive, while the pilot

router mismatch is very small relative to the mismatch incurred by committing to a single global model.

Table 1: Uniform-prior decomposition terms at $\tau = 0.05$.

Quantity	Point estimate	95% bootstrap CI
Cost term $\log \kappa_0 / \kappa$	-1.424	$[-1.497, -1.318]$
Competence term ϕ	6.817	$[6.561, 6.908]$
Pilot information gain G_{pilot}	0.510	$[0.088, 0.648]$
Holdout information gain G_{holdout}	0.660	$[0.635, 0.686]$
Pilot mismatch $\varepsilon_{\text{pilot}}$	0.033	$[0.0003, 0.513]$
Single-model mismatch $\varepsilon_{\text{single}}$	6.959	$[6.688, 7.016]$
Pilot-router holdout objective	4.894	$[4.523, 8.472]$
Oracle-router holdout objective	4.541	$[4.505, 4.625]$
Single-best-model holdout objective	10.696	$[5.189, 10.778]$

Two conclusions follow directly.

First, the four quantities are diagnostic on a real benchmark. Cost, competence, information gain, and mismatch are all measurable with numerically stable estimates. In particular, the holdout information term and the single-model mismatch term are already tight enough to support a strong diagnostic claim.

Second, the pilot router is already useful. The pilot router is not perfect, but it is close to the oracle and decisively better than a single global model. This is the practical model-selection result that benchmark racing misses: even before the matrix is fully powered, the estimated task-mode structure already points to a better deployment choice.

4.4 Cell-Level Evidence

Table 2 shows the current cell-level success rates and Wilson intervals for the pairwise study. The decisive cells are already well separated:

- holdout `range_local_scans` / spark: 10/10
- holdout `topk_stress` / spark: 0/10
- holdout `range_local_scans` / codex: 5/6
- holdout `topk_stress` / codex: 2/2

The codex `topk_stress` cells remain the weakest part of the current matrix, so the correct claim is still *partial real-benchmark validation*, not a fully closed large-sample theorem. But the sign pattern required by the theory is already clear and stable.

Table 2: Current pairwise cell counts at $\tau = 0.05$.

Split	Task mode	Model	Successes / trials	Wilson interval
Pilot	range_local_scans	spark	8/10	[0.490, 0.943]
Pilot	range_local_scans	codex	6/6	[0.610, 1.000]
Pilot	topk_stress	spark	1/10	[0.018, 0.404]
Pilot	topk_stress	codex	2/2	[0.342, 1.000]
Holdout	range_local_scans	spark	10/10	[0.722, 1.000]
Holdout	range_local_scans	codex	5/6	[0.436, 0.970]
Holdout	topk_stress	spark	0/10	[0.000, 0.278]
Holdout	topk_stress	codex	2/2	[0.342, 1.000]

4.5 Implications for the Claim

The strongest defensible claim supported by the current evidence is:

On a real verifiable oracle-family benchmark, the four-term decomposition is diagnostic and practically informative. It distinguishes routing-trivial and routing-non-trivial regimes, measures the relative contributions of cost, competence, information, and mismatch, and identifies a pilot router that is substantially closer to the oracle than any single global model.

This is stronger than “the terms exist” and narrower than “the theory is universally validated.” It is the right level of empirical claim for the current state of the data.

5 Multi-Model Expansion

The repository now also supports a five-model campaign planner and NeurIPS-style visual stack. Operationally, the reliable five-model set is:

- `claude-haiku`
- `gpt-5.3-codex-spark`
- `gpt-5.3-codex`
- `gpt-5.4-mini`
- `gpt-5.4`

This multiclass expansion is not yet statistically powered, so it remains supporting infrastructure rather than main-paper evidence. The key value for the paper is that the decomposition framework now has a clean path from two-model diagnostic validation to a broader model menu without changing the underlying formalism.

6 Limitations

The updated empirical section improves substantially on the original synthetic state, but four limitations remain.

1. Task modes are oracle workload families, not fully discovered latent modes.
2. The real-benchmark validation is still strongest for the two-model pairwise case; the multiclass campaign is not yet fully powered.
3. Some codex `topk_stress` cells are under-sampled, so uncertainty remains asymmetric across cells.
4. The one-step attempt setting is the right theorem-facing regime for this paper, but it does not exhaust the longer-horizon agentic optimization question.

These are scoping limitations, not contradictions of the theory. They define what the current results establish and what the next experimental budget should target.

7 Conclusion

The decomposition remains useful for the same reason as in the original manuscript: it converts a vague deployment question into separately measurable diagnostics. What has changed is the empirical footing. The repository now contains a real-benchmark validation in which the decomposition:

1. identifies a true threshold transition between trivial and non-trivial routing regimes;
2. measures the four terms directly with uncertainty intervals;
3. exposes a large gap between a pilot router and a single global model;
4. and explains *why* the preferred design changes with task mode.

The most important next step is not another synthetic experiment. It is to finish powering the remaining weak cells and then scale the same oracle-family protocol to the reliable five-model set. But even before that larger study is complete, the current evidence already supports the refined central claim: decomposable performance accounting is a practical diagnostic for model selection on real verifiable agentic tasks.

A Proof of the Decomposition Theorem

A.1 Proof of Theorem 1

Proof. Under Assumption 2.1, conditional on mode s and routing allocation q_D , the packed-family parametrization gives

$$\mathbb{E}[-\log T(M, q) \mid S = s, D] = c + \log q_D(s) + \log p(s, M),$$

where c is independent of s , M , and the routing choice.

Taking expectation over (S, D) and adding the per-attempt cost term gives

$$\begin{aligned} \mathcal{J}(M, q) &= -\log \kappa(M) + \mathbb{E}[-\log T(M, q)] \\ &= -\log \kappa(M) + c + \sum_s \pi_s \log p(s, M) + \mathbb{E}_{S,D}[\log q_D(S)]. \end{aligned} \tag{4}$$

It therefore remains to rewrite the routing term $\mathbb{E}_{S,D}[\log q_D(S)]$. By definition,

$$I_\pi(S; D_M) = \mathbb{E}_{S,D} \left[\log \frac{\pi_D^M(S)}{\pi(S)} \right], \quad (5)$$

$$\varepsilon(M, q) = \mathbb{E}_D [\text{KL}(\pi_D^M \parallel q_D)] = \mathbb{E}_{S,D} \left[\log \frac{\pi_D^M(S)}{q_D(S)} \right]. \quad (6)$$

Subtracting (6) from (5) yields

$$I_\pi(S; D_M) - \varepsilon(M, q) = \mathbb{E}_{S,D} \left[\log \frac{q_D(S)}{\pi(S)} \right] = \mathbb{E}_{S,D} [\log q_D(S)] + H(\pi),$$

because $-\mathbb{E}_{S,D}[\log \pi(S)] = -\sum_s \pi_s \log \pi_s = H(\pi)$. Hence

$$\mathbb{E}_{S,D}[\log q_D(S)] = -H(\pi) + I_\pi(S; D_M) - \varepsilon(M, q). \quad (7)$$

Substituting (7) into (4) gives

$$\mathcal{J}(M, q) = C - \log \kappa(M) + \sum_s \pi_s \log p(s, M) + I_\pi(S; D_M) - \varepsilon(M, q),$$

where $C = c - H(\pi)$ is independent of the design. Writing the same expression for the reference design (M_0, q_0) and subtracting, the constant cancels and we obtain

$$\begin{aligned} \Delta(M_0, q_0 \rightarrow M, q) &= \log \frac{\kappa_0}{\kappa(M)} + \sum_s \pi_s \log \frac{p(s, M)}{p(s, M_0)} \\ &\quad + (I_\pi(S; D_M) - I_\pi(S; D_{M_0})) - (\varepsilon(M, q) - \varepsilon(M_0, q_0)), \end{aligned}$$

which is exactly (1). $\square$

A.2 Proof of Theorem 2

Proof. Apply Theorem 1 to a reference design that uses no side information. If D_{M_0} is degenerate, then $P(S = s \mid D_{M_0}) = \pi_s$ for every mode, so

$$I_\pi(S; D_{M_0}) = 0.$$

If the reference routing also equals the prior, $q_0(s) = \pi_s$, then

$$\varepsilon(M_0, q_0) = \mathbb{E}_{D_{M_0}} \left[\text{KL}(\pi_D^{M_0} \parallel q_{0,D}) \right] = \text{KL}(\pi \parallel \pi) = 0.$$

Substituting these two identities into (1) yields (2). $\square$

B Proof of the Routing-Triviality Criterion

Proposition 1. *Define the per-mode cost-adjusted score*

$$\rho(s, M) = -\log p(s, M) + \log \kappa(M).$$

If for all modes s and all model pairs M, M' one has

$$|\log p(s, M) - \log p(s, M')| < |\log \kappa(M) - \log \kappa(M')|,$$

then routing is trivial: the cheapest model minimizes $\rho(s, \cdot)$ on every mode. Conversely, routing is non-trivial only if there exist modes s_1, s_2 and models M_1, M_2 such that

$$\rho(s_1, M_1) < \rho(s_1, M_2) \quad \text{but} \quad \rho(s_2, M_1) > \rho(s_2, M_2).$$

Proof. Let M_c be the cheapest model, so $\log \kappa(M) - \log \kappa(M_c) \geq 0$ for all M . For any mode s and any other model M' ,

$$\rho(s, M') - \rho(s, M_c) = (\log p(s, M_c) - \log p(s, M')) + (\log \kappa(M') - \log \kappa(M_c)).$$

By assumption,

$$|\log p(s, M_c) - \log p(s, M')| < \log \kappa(M') - \log \kappa(M_c),$$

so the second term dominates the first and $\rho(s, M') - \rho(s, M_c) > 0$. Therefore M_c minimizes $\rho(s, \cdot)$ for every mode, hence routing is trivial.

For the converse, if no crossing exists, then one model has weakly smaller $\rho(s, \cdot)$ on every mode and routing again collapses to a single global choice. Therefore non-trivial routing requires at least one pair of modes on which the ordering of $\rho(s, \cdot)$ reverses across models. $\square$

References

- [1] Alessandro Achille and Stefano Soatto. AI agents as universal task solvers: It’s all about time. *Entropy*, 28(3):332, 2026.
- [2] Pierfrancesco Beneventano and Tomaso Poggio. A mathematical framework for building agent systems: task-first design, equivalence, and model choice under deployment budgets. Under review, 2026.
- [3] Bradley Brown, Jordan Juravsky, Ryan Ehrlich, Ronald Clark, Quoc V. Le, Christopher Ré, and Azalia Mirhoseini. Large language monkeys: Scaling inference compute with repeated sampling. arXiv:2407.21787, 2024.
- [4] Charlie Victor Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling LLM test-time compute optimally can be more effective than scaling model parameters. In *ICLR*, 2025.
- [5] Yangzhen Wu, Zhiqing Sun, Shanda Li, Sean Welleck, and Yiming Yang. Inference scaling laws: An empirical analysis of compute-optimal inference for problem-solving with language models. In *ICLR*, 2025.
- [6] Lingjiao Chen, Matei Zaharia, and James Zou. FrugalGPT: How to use large language models while reducing cost and improving performance. arXiv:2305.05176, 2023.
- [7] Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez, M. Waleed Kadous, and Ion Stoica. RouteLLM: Learning to route LLMs with preference data. arXiv:2406.18665, 2024.